§20 of the isogeny-fibers note: round 7, verdicts and corrections

Date 2026-11-01 (branch `research/note-s20-round-7`; the revised §19 from #3072 is on `main`). **Type** documentation of existing results; no experiment was run for this report and no number below is newly computed unless marked *derived*. Toy and synthetic instances only; no sub-rho claim and no claim about a deployed curve.

Question. What did the eight round-7 leads (R7.1–R7.8) find against their hypotheses and falsifiers, and which statements of §19 must change in light of them?

Method. Read each lead's source record (R7.5: the two `ERRATUM_v2.md` files; R7.8: the pull-request description, because it has no `RESULTS.md`), checked the branch heads against the pull-request records (all eight match), and transcribed hypothesis, falsifier, verdict and numbers with the sources' intervals. Five of the eight pull requests are merged as of 2026-11-01 04:11 UTC (R7.1 #3107, R7.3 #3105, R7.5 #3109, R7.7 #3113, R7.8 #3125, merged at 03:58 UTC); the other three are **open and unmerged**: R7.2 #3124, R7.4 #3092, R7.6 #3128. The §19 update #3072 merged on 2026-11-01; #3074 remains open. One check was re-run for this report: both `recount_v2.py` scripts of R7.5 on the committed records of `main`; their output equals the committed `recount_v2.txt`.

The verdict figure (Figure 1) is on the last page (landscape).

Findings, including negative and unresolved ones

- **Falsifier fired / refuted.** R7.2: no toy instance is weak; index calculus costs 1.91 [1.72, 2.11] to 131 [123, 141] times the best generic method on 30 instances (N = 12 to 28), every interval above 1. R7.4: the binary-Edwards pair-table falsifier as worded fires at n ≤ 12 (39 of 40 cells exclude 1) and no gain is claimed. R7.6: the path-finding exponent is 1.13 [1.06, 1.21], so $O\left(\sqrt{C}\right)$ is refuted for breadth-first bidirectional search. R7.7: the a = 1 premise is false ($E/4E \cong \mathbb{Z}/2$).
- **Mixed.** R7.1: the ordering falsifier fires in one accounting (cold, ascending: ℓ = 89, 109, 131, near ties of 1.6%, 0.3%, 0.4%), the exponent falsifier does not; the exponent stays 1.919, insufficient to show a value below 2. R7.3: null in iterations and yield after the $|\mathrm{Aut}|$ baseline; one inferior algorithm costs 1.46× [1.44, 1.48] more on the floor in multiplications. R7.6: pooling gain exponent 0.43 to 0.52 against ideal 0.489, but below $\sqrt{C}$ by more than its interval from C = 32. R7.7: the 4E criterion and odd-coset labels hold for every (a, b) with Tr(a) = 0, constancy of the second bit is measured not proved, and the constant-time bar is still missed there.
- **Not evaluated or unread.** King §5 and the Eagle–Galbraith–Ong cofactor-4 test: access blocked, still unread (R7.4). No Gröbner/F4 oracle on Edwards summation polynomials. The extrapolated crossing at N ≈ 32 (R7.2) was not built.
- **Corrections.** R6.7: 11 / 9 / 11 of 18 intervals contain 1 (not 12), and the m = 3 pricing shift is 0.0151 (not 0.011); R6.9: 7 of 34 runs are more than 10× from 1/g! (not 6). No verdict changes (R7.5).
- **Library implementation follow-up (merged #3125).** `chained-x` is registered in `ic bench`, and the end-to-end procedure-2 test is added. The PR description reports an alternate-base comparison; those `chained-x` measurements are not bit-identical to R6.1′s because the base differs. It adds no separate research hypothesis or verdict.

What changed in the note (research note, not a result)

New §20 (one subsection per lead, synthesis table, settled and open lists, next leads, edit list). Edits to earlier sections: header and §0 item 10; §19 status paragraph; §19.5, §19.11 (the 0.011 sentence and the unscoped inversion share), §19.13, §19.16 rows, §19.17 open lines (each now names its round-7 result and its pull-request status), §19.18, §19.19. The "awaits correction" lines were kept in substance because R7.2, R7.4 and R7.6 remain open; the R7.5 errata (#3109), R7.8 library follow-up (#3125) and #3072 merged. §17.3 and all frozen results are untouched; measured values of earlier sections are unchanged.

Graphs checked and updated

`round7/s20_summary/verdicts.json` and its rendered `verdicts.svg` were updated to mark 3125 merged; `report.pdf` was regenerated and inspected with the updated figure. `round6/s19_summary/` covers rounds 5 and 6 and is unchanged; each round-7 lead's own graphs are linked from §20. The scoreboard, ledger and `benchmarks/` graphs were not searched on 2026-11-01 (the 2026-10-24 search of §19 was not repeated); no ledger or scoreboard value changes. No PNG render is committed (the repository's artifact policy sends a PNG over 64 KiB to a store this session cannot write).

Limits and unresolved checks

Verdict classes compress mixed outcomes into one column; the wording is in §20.9. Merge status was checked on 2026-11-01 at 04:11 UTC: R7.1, R7.3, R7.5, R7.7 and R7.8 merged; R7.2, R7.4 and R7.6 remain open. CI for those remaining open research PRs was not audited in this documentation update. Conductor task T-1211 reserves the note and round-7 summary directory. The curve-name linter was not run (docs only; the note names no curve slug that §19 did not already carry).

Commands

`python3 plot.py` with Matplotlib (reads `verdicts.json`, writes `verdicts.svg`); `typst.compile('report.typ', output='report.pdf')` in the isolated Typst Python environment.



Hollow marker: pull request open and UNMERGED; filled: merged. Status checked 2026-11-01 04:11 UTC; five merged (including R7.8 #3125 at 03:58 UTC), three open. One verdict class per lead; see section 20.9.
R7.8 has no separate research verdict. R7.1: the ordering part of the falsifier fires in one accounting, the exponent part does not. Toy instances only. Source: verdicts.json.

Figure 1: Verdict class of each round-7 lead (transcribed; data `verdicts.json`, script `plot.py`). Hollow markers: pull request open and unmerged; filled: merged as checked on 2026-11-01 04:11 UTC (five merged, three open). R7.1: ordering falsifier fires in one accounting, exponent falsifier does not. R7.8 has no separate research verdict.